Dimitri Chrysafis

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Education

University of Wisconsin–Madison Graduating May 2027
Bachelor of Science in Computer Science
Relevant Coursework: COMP SCI 577 (Algorithms), COMP SCI 524 (Optimization), MATH 435 (Cryptography); A in each

Technical Skills

Languages: Python, C++, Haskell, Julia, Go, JavaScript
ML/GPU: PyTorch, CatBoost, WebGPU/WGSL, WebGL/GLSL, Apple Metal, inference, quantization, model serving

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Experience

Quantum Circuit Routing [Final Report] **COMPSCI 524 Project** Jul 2026

- Built an exact Julia/JuMP/HiGHS MILP for routing quantum circuits on sparse superconducting hardware
- Solved 24 instances to proven optimality; grid3x3 lowered the objective **56.1%** versus heavy-7

Projects

Qwen3.6 NVMe Inference Engine

- Developing a quantized, disk-backed Qwen3.6-35B-A3B format targeting a reduction from **about 25 GB** for standard Q4 inference to **about 12 GB** resident memory by streaming router-selected MoE experts from NVMe
- Building a runtime with memory-mapped weights, bounded expert caching, and asynchronous prefetching under configurable memory budgets; measuring cache hit rate, exact-logit agreement, throughput, latency, and minimum memory

Interactive 3D Fluid Simulator (MLS-MPM) [Demo] [GitHub]

- Built a browser-based 3D MLS-MPM simulator in JavaScript/WebGPU, keeping particle and grid state in GPU storage buffers and simulating **400,000 particles** with staged compute kernels
- Implemented fixed-point atomic accumulation for particle-to-grid transfers plus interactive dam-break, piston, particle-injection, and orbit-camera controls

Tennis Match Prediction System

- Built a feature pipeline over **92,000 ATP matches (1982–2024)** with player and surface-specific ELO, head-to-head, recent-form, and schedule-fatigue features
- Trained a CatBoost ensemble to predict winners; achieved **83.7% accuracy** and outperformed a baseline model

GPU Fractal Rendering Engine [GitHub] [Blog]

- Built GPU fractal renderers in WebGL2/GLSL and Apple Metal for Newton, Kleinian, and Julia sets with interactive pan/zoom and shader-based iteration
- Implemented a Metal compute pipeline for **6,000×6,000** Julia frames with adaptive iteration counts and GPU-parallel image generation

GPU-Accelerated Sphere Packing in Arbitrary Meshes [GitHub] [Blog]

- Built a stochastic tangent-sphere packing solver for arbitrary OBJ meshes, enforcing interior, surface-clearance, and pairwise non-overlap constraints
- Accelerated ray-mesh containment, nearest-surface queries, and candidate search with batched PyTorch tensors on MPS/CUDA; visualized results in Open3D